Ahmad Hakim

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PROFILE

Motivated and detail-oriented Computer Science student with strong analytical skills, programming experience, and a passion for scientific research and advanced technology. Known for academic excellence, fast learning, and dedication. Seeking a training opportunity at CERN to gain hands-on experience in computing, data analysis, and large-scale scientific systems.

EDUCATION

Effat University
Bachelors, Computer Science *Jeddah, KSA*
- Dean's List 2024-2025
- GPA: 4/4

Al Zahrawi High School
- Graduated with 99.9%

SKILLS

- **Programming & Software Development -** C++, Java, Python, PHP, JavaScript, HTML, CSS, Object-Oriented Programming (OOP), Data Structures & Algorithms, API integration (TMDb, OpenAI, custom REST APIs)

- **Web Development -** Full-stack development using PHP, MySQL, JS. Database design & optimization (ERDs, relationships, normalization). Creating responsive UI/UX design (Bootstrap, custom CSS)

- **AI, Machine Learning & Data -** Application of Physics-Informed Neural Networks (PINN) for scientific modeling. Understanding of regression models (Intel baseline, Random Forest, FFNN, KNN). Data cleansing, analysis, and visualization (Pandas, Matplotlib, CSV handling). Model evaluation (RMSE, MSE, training/testing splits).

- **Mathematics & Problem-Solving -** Strong proficiency in Discrete Mathematics, Calculus I/II, Linear Algebra, and Statistics I/II. Experienced in applying eigenvalues/eigenvectors, vector spaces, partitions, and other core mathematical concepts. Excellent analytical thinking with clear, step-by-step problem solving. Experienced in tutoring university-level math.

Soft Skills: Fast learner with strong academic excellence, good tutoring capability, Attention to detail, Verbal and Written Communication, Adaptibility, Interpersonal Communication.

ACADEMIC WORK & PROJECTS

Research Assistant – Red LED Lifetime Modeling Project with Dr. Omar Kittaneh & Dr. Mohammad

Abdulmajid **(In Progress)**

- Working on a research paper investigating Red LED lifetime prediction using machine learning and physics-informed methods.
- Implementing and comparing models such as Physics-Informed Neural Networks (PINN), Random Forest, and baseline Intel models.
- Conducting scenario-based testing, RMSE evaluation, and reliability analysis.
- Preparing experimental results and figures for academic publication.

CineVault Movie Analysis Website **March 2025**

- Full-stack PHP & SQL web application using The Movie Database (TMDB) API.
- Includes login and user authentication
- Utilizes the TMDB API to showcase movie/TV shows with their ratings and related information

Legal Retrieval-Augmented Generation (RAG) System – KSA Law **(In Progress)**

- Developing a LLaMA-based RAG system to analyze and answer legal questions related to Saudi Arabian law.
- Building a pipeline including document ingestion, chunking, embeddings, vector storage, and retrieval.
- Implementing context-aware RAG for lawsuit classification, article lookup, and legal reasoning.
- Working with Python, LangChain, LLaMA, and vector databases (FAISS/Chroma).
- Designed for legal research assistance and query answering.

VOLUNTEER EXPERIENCE

Ideal Innovations Office Furniture **Summer 2025**

▪ Provided on-site IT support and receipt handling.

▪ Helped with the sales team and customer support.

Peer Tutoring **January 2024 - Present**

▪ An active member in the peer tutoring program offered in the university, helping students with mathematics-based courses.

CERTIFICATIONS

▪ **IBM - Gen AI Foundational Models for NLP & Language** **Oct. 2025**

▪ **Udemy - Probability / Stats: The Foundation of Machine Learning** **July 2025**

▪ **IBM - Fundamentals of AI Agents Using RAG and LangChain** **Oct. 2025**

▪ **IBM - Python for Data Science, AI & Development** **Oct. 2025**